

How we're addressing comments

Reviewer 1:

1. For hierarchicalRM we use SceneDescriptor + StateDescriptor to get privileged state info from an image. However, in the baseline, we show REPLAN and ReplanVLM don't solve tasks but we do. They're not sure if the performance gain is due to our pipeline and hierarchy or the fact that we use privileged state info but the baselines have to rely on raw images.

Resolution – They want us to run baselines with same scene + state desc and isolate to compare just core planning pipeline giving both baselines our privileged state info to see how well they perform now.

2. The authors claim the H1 Action Generator composes sequences that yield "exactly one relational change." But the operational details are completely missing. How exactly are preconditions and effects declared, validated, and guaranteed? Are these schemas generated on the fly per scene, or pulled from hardcoded templates? The paper needs a library table detailing the exact preconditions and effects for the H1 and H2 actions used across the tasks.

Resolution – We need to define what we meant by Transfer, if its hardcoded or not. We need a library table showing the preconditions and the effects that H1/H2 actions yield. I imagine given we're going to dynamically construct them, the preconditions are derived from the task constraints. Let's make this explicit. Furthermore, the validity of these action sequence is going to be determined by the Inner Bot verifying the plan + Decision Bot using these in execution. I think the new dynamical hierarchy approach shouldn't have this issue.

3. The Multi-Ring Transfer benchmark is fundamentally a discrete, fully observable, constraint-rich domain once the scene is parsed into a symbolic state. Given these characteristics, classical symbolic planners or task-and-motion planning methods could be highly competitive. The absence of a classical baseline, such as a PDDL solver utilizing the exact same predicates and constraints derived from the JSON state, limits the ability to properly contextualize the specific advantages of using reasoning models for this class of problems. Would it be possible to include a classical planner baseline on tasks T1 through T10 to demonstrate where the reasoning-based approach specifically excels compared to traditional search methods?

Resolution – We need a classical PDDL planner for hanoi tower and show how far it succeeds compared to our solution. For other benchmark, maybe Lexicon-Sokoban dataset also has a PDDL solver but I'm not fully sure how we can use this given the task

complexity is defined based on what the PDDL solver determines on how easy it is to solve the task. Perhaps, we can make the task complex and give it to our pipeline + a classical baseline planner to show how and where reasoning models help.

This won't apply for the robotics Simulator experiments we do but could be added if required to address the puzzle task benchmarks.

4. For the real-robot dVRK execution, the Scene Descriptor implementation is under-specified — unclear if it's VLM-based parsing or simple color/geometry heuristics. Requests per-object perception accuracy and primary failure modes for reproducibility.

Resolution – We need to explain in more clarity how does scene descriptor on real experiment setup vary than what we do for simulation.

5. The Outer Bot classifies errors as "Recoverable" or "Fatal" but the criteria for this split are undefined. Asks whether these are predicate-based thresholds or free-form judgments by the reasoning model.

Resolution – Make the distinction clear on what each type of error means and how we decided this. I believe today, this is only said in the Algorithm and not in much detail. A brief paragraph making this explicit will help.

6. No computational run-time statistics are reported, despite surgical autonomy being latency-sensitive. Requests mean/std of planning and verification latencies per subtask and per episode, for both simulation and real-robot runs.

Resolution – Run times in real robot experiments is requested by reviewer to show how much latency our pipeline introduces for simulation and real robot runs per subtask execution per task. This detail can be summarized in a table to save space given 8 page constraints.

7. The complexity score (Eq. 1) uses ad-hoc weights (6, 4, 1.5); the "up to 3x more complex" claim depends entirely on these. Requests a sensitivity analysis over the weights. Also, since real-robot results use only 5 runs/task, requests variance/error bars across seeds for success rate, replan counts, and token usage.

Resolution – This should be addressed given we don't define complexity score anymore as TOH and other baseline tasks are well defined problems with complexity known and mapped. As for statistics and runs/task, we can add variance/std to tables with +/- indications. Furthermore, we could improve the graphs and make the error bars/lines very explicit to call this out for various statistics reported. This must be done across various seeds. Although, I'm not sure if I've read papers do it with such rigor to change seeds and do it.

Reviewer 2:

1. The claim that prior peg-transfer automation "has been limited to single-step maneuvers" unfairly criticizes prior work — peg transfer is an FLS task specifically designed to train fine motor skills, not cognitive reasoning, so criticizing it for focusing on low-level control is misplaced.

Resolution – We need to relax the prior works as in be very careful with why we say they dont work. A controlled experiment on giving same privileged state info like how Reviewer 1 said could be one way. Alternatively, we could just test them on newer benchmarks. Our framing needs to move out of the surgical robotics and should be general robotics as a big concern is lack of showing how peg transfer generalizes to other robotics tasks. They said primitives task that literature review papers do is what generalizes to tissue manipulation for instance.

2. Multi-Ring Transfer, while a good testbed for long-horizon planning, is essentially a dVRK version of Tower of Hanoi / tabletop block-stacking puzzles. Requests better articulation of the pathway from this structured puzzle environment to real, unstructured surgical task planning.

Resolution – Similar to above, moving out of purely surgical domain focus and general robotics would help. In addition, extending our experiments to flat hanoi + sokoban puzzle task to convince long horizon + Libero dataset sim experiment + real robot flat hanoi would close the story well for why long horizon and its application in robotics. Benchmarks on real robot will be key because they may argue puzzle task is one thing but do we notice degradation by prior works in robotics tasks as well? If we show our pipeline working then thats huge and solid result. The puzzle simulation is motivation + reason to show on real robot. Real test is the robotic simulation tasks.

3. The paper critiques prior hierarchical methods for requiring "extensive domain knowledge" and "intricate reward engineering," but HierarchicalRM itself relies on hand-crafted primitives, an explicit JSON schema, and hard-coded constraint programming — this just shifts the domain-knowledge burden from reward engineering to prompt/schema engineering. Requests softened flexibility claims relative to prior work.

Resolution – Hand crafted primitives issue would be resolved with using pre-trained VLA as we're moving away from primitives engineered ourselves so thats our differentiation. The explicit JSON schema is a fair argument so we'd need to show why not direct image – I believe this was something we discussed as do we need scene + state descriptor or is language task instruction + constraints described + input image enough? If the latter works, this would also address the hardcoded constraint programming and JSON schema comments as their comment on prompt/schema engineering is a valid argument. **Perhaps something to run few tests on ourselves to verify if our pipeline can be simplified in certain modules? – We will remove state**

descriptor and focus on H1/H2 actions/dynamical hierarchy.

4. The claimed VLM limitation of lacking "explicit state" is addressed via JSON schema, and ablations show removing descriptors hurts performance — but there's no comparison against state-of-the-art implicitly-learned representations to prove the explicit JSON approach is actually superior. (not an issue as without scene its implicit model vision capabilities)

Resolution – This could perhaps be made explicit if we show baselines that use images as inputs for certain baselines we compare against. Then again, it might be difficult to isolate and show JSON is superior or it could be their pipeline doesn't work well for these experimental tasks we do. Similar to #3 point, I would like to see if perhaps we can remove JSON state to images or find a reason to make the argument to why we need JSON. **I need some input here.**

5. Table I lists "Generate reusable action" as a unique HierarchicalRM feature, but without a baseline isolating dynamic action generation vs. simply having a predefined hierarchical library, it's unclear if this generation mechanism is a real improvement over prior VLM approaches. (lets remove reusable comments/contribution)

Resolution – This goes to the point about reusing skills/actions generated previously. Technically, we never did this hence why they brought it up. Like is replanning from start better or same as reusing prior skills generated? That's what we need to answer. I think they mean can we show our pipeline reuses hierarchies and gets better performance compared to having predefined library of actions that we give to LLM and see if it reuses those/uses those to get the task done? I THINK an argument in favor to avoid doing this is having to hand engineer the primitives everytime so I'm not sure why they brought this up.

6. The dynamic grasp logic (temporarily disabling ring rigidbody on grasp to prevent slip-outs) bypasses real physical grasping complexity. Requests discussion of how this physics abstraction affects the sim-to-real gap and robustness of the low-level motor module.

Resolution – Yeah, that's fair. Given we're going to not use Unity for updated paper, this should be resolved as we use Libero SIM and grasping in real works fine based on what we have in code today. This was a problem specifically in Unity Simulator.

7. **Video comment:** The supplementary video is too long/detailed, mostly text slides and disjointed clips without voiceover. Recommends simplifying to focus on high-level contributions with clear, narrated footage of simulation and real dVRK experiments.

Resolution – Trim the video, keep it brief and add voiceover if necessary for sim and real dvrk experiment.

Reviewer 3 (PDF):

1. The paper frames the work as advancing "autonomous surgical planning," but the benchmark is a colored ring rearrangement task on pegs with no deformable tissue, tool-tissue interaction, bleeding/occlusion, bimanual manipulation, suturing, or anatomy-specific constraints. Suggests softening claims like "advancing autonomous surgical planning."

Resolution – Similar to reviewer 2 comment and my opinion is, we move into general robotics and away from surgical robotics claims and arguments.

2. The complexity score is customized and unverified — the paper admits the weights are chosen to "keep terms comparable" but this is arbitrary, and the headline "3x more complex" claim depends heavily on it. Suggests empirical justification (fitting weights to baseline failure rates), a sensitivity analysis over weights, or a softened claim (e.g., "solves tasks with substantially higher CR and PR than baselines").

Resolution – This problem arose due to the open-endedness of the TOH setup we had with custom constraints. By using predefined puzzles that are known to be hard for LLMs with prior work results, this review should be addressed.

3. Table III lists the models used, raising reproducibility concerns: models may not be publicly accessible or may change versions; the submission date vs. referenced model names raises questions; if models are API-hidden, more detail is needed (temperatures, prompts, number of samples, retry policy, context length, exact model identifiers). Requests a reproducibility appendix with all prompts, decoding parameters, API versions/snapshots, and failure-handling details.

Resolution – The various parameters for configs to our LLMs needs to be provided part of our anonymized codebase/website that we publish or the GitHub. We can reference as one line and have core prompts and details on there given the 8 page constraint limit.

Furthermore, showing open source should help since they're never deprecated which is a concern reviewer has for closed source models (like o1 not being available via API 2 years from today). We should definitely focus on open source models more if not just as much as closed source. Open source also helps with our focus and claim on anyone being able to run it on smaller reasoning models or on your machines without being gated by costly API from these model providers.

Any config/parameter like temp, context window, reasoning effort, etc even default should be made explicit for open source and closed source.

We should use precise model identifiers like `gpt-5.6-luna-2026-02-15` or whatever the model provider website says. Similarly, `qwen3-8B`, `gpt-oss-20B` for

instance.

4. It's unclear whether REPLAN and ReplanVLM baselines were adapted with equal care — if they operate closer to raw VLM planning, the comparison may conflate "reasoning model superiority" with "better scaffolding," making it unclear whether the hierarchical structure or the RMs themselves are the key factor.

Resolution – Clarity on how the baselines were adapted to accurately reflect their usage in our experiments (outside of what the original paper reported) is necessary.

Furthermore, they're not convinced with hierarchy showing improvements or frontier models reasoning being the factor in better performance. This can be addressed by using weaker reasoning models and more specifically open sources low parameter models. **This is a required baseline and model config to show with.**

5. The claim that reasoning depth correlates with task success is only observational (Figure 7); token count may not equal reasoning depth — shorter traces could result from API behavior, prompt structure, model refusal/failure modes, or hidden implementation details.

Resolution – This should partly be addressed as if we're using known benchmarks and defined complexity, like puzzles flat-hanoi, sokoban that have a natural scaling and formulation for complexity that is proven mathematically, then the claim on token count with reasoning depth is valid in my opinion because it is mathematically given this task is more complex than the previous data point task reported on the graph and the observed behaviour is expected (eg: 5 ring vs 6 ring task).|

Now, this is important to note since flat hanoi for any hoop has various intermediate configs so we need to clearly identify of the possible configs, which ones are known to be hard mathematically or otherwise and use those as our benchmark for hard config, medium or easy config when reporting data.

6. Real robot experiments use simplified color/position perception, one PSM, 3D-printed rigid rings, and conservative primitives, and don't cover T8–T10 (where simulation results are most interesting). Requests an explicit explanation for why T8–T10 weren't run on hardware, and quantitative failure category reporting (missed grasp, slip, perception error, unreachable ring, constraint violation, planner error, etc.).

Resolution – Need input from Amey – Just show on few flat hanoi configs for real. Keep as is for now

7. Overall presentation is confusing at times — inconsistent/confusing terminology, notations, and definitions — and some text reads as heavily AI-polished writing style.

Resolution – Need to clean up AI written slops and we need to make terminologies, notations and other definitions clear and reference them where applicable or write them

better perhaps. The review didn't say anything explicit on what exactly but a note for us to write it in more readable manner.

Other Misc Comments from Annotated PDF:

1. **How do we differentiate RM, VLM, LLM** – I guess based on SOTA model providers and benchmarks established and the roles of each type? Like how explain this?
 2. **Surgical task/clinical value comments** – Ignored as we're moving out of surgical robotics right and none of our experiments are "surgical".
 3. **Subtitle headings indicate AI-Generated** – "Benchmarks for RAS", "General Knowledge (GK)" and others needs to be human written/explicitly cleaned.
 4. **Only since benchmark on multi peg transfer** – Addressed by new benchmarks and experiments in revised paper.
 5. **Table 1 seems too suspicious on claims made before experiments** – Suggested to prove with experiments and rigor and move to discussion/experiments section.
 6. **Confusing notations on H1/H2 and subscripts** – Clean up and use concise language and notations to explain the math/algorithm. The notations used in the algorithm are also not clear and confusing. Suggested to simplify and use common notations from literature if they exist. Also, explain these in the text and not just in algorithm section.
 7. **Confusion on if H2 generator is task agnostic generation** – Confused on how robust is the pipeline and its correctness. Asked for clarity on task specific and task agnostic components.
 8. Asked to define experiment protocol in 'Experiments Setup' section of paper instead of report – Page 6/8. See below comment
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frequency, and total SC. Unlike simulation RC, this metric counts only replanning events triggered by the Outer Bot following execution-time verification failures.

11.2 ON 15j.

2) *Error recovery*: Execution errors such as missed grasps or rings slipping after pickup often yield *Recoverable* states,

9. Concerns on why certain experiments/ablations are only on one baseline and some others use another. All of them should use all baselines for standardization.
10. Need concrete examples of H1/H2 actions defined as we reuse these terms in paper esp in experiments/results section but its not very clear.

11. Some graphs aren't very clear on which subgraph/plot is ChatGPT and which one is Claude – Make graphs explicit.

Table1: SR

	Lexicon	Flathanoi	XYZ	
Gpt5.6 luna				
With hierarchy				

Table 2:

	No of replans	H1 calls
Lexicon		